SECURELINK VPN LAB

Research Project | Secure VPN for Remote Access

Monitoring & Logging Plan

Establishing a Secure VPN for Remote Access

Deliverable D4 of 6

Organisation	Research project (lab build)
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Version	1.0 (final)
Basis	Lab build on VMware; evidence IDs [E01]-[E38] refer to the evidence screenshot set
Status	Lab documentation. Not certified and not presented as production-ready.

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1 Objectives

The aim is to track VPN usage, keep an audit trail and support incident analysis. All server times are UTC; dates are as shown in the cited screenshot. ELK, Nagios and Zabbix were not deployed, so monitoring in this lab is log-based.

2 Log sources

Source	What it records	Evidence
OpenVPN status file	Currently connected client	[E19]
journald / OpenVPN logs	Connections, TLS and auth results	[E09][E24]
auth.log (PAM)	TOTP accepted and invalid code	[E09]
Kernel log (iptables)	VPN-DENY drops of non-permitted traffic	[E11]
charon (strongSwan)	IPsec tunnel events	[E01][E28]
fail2ban sshd jail	SSH failures, ban and unban	[E17]
sshd / sudo	Logins and privileged commands	[E27][E28]

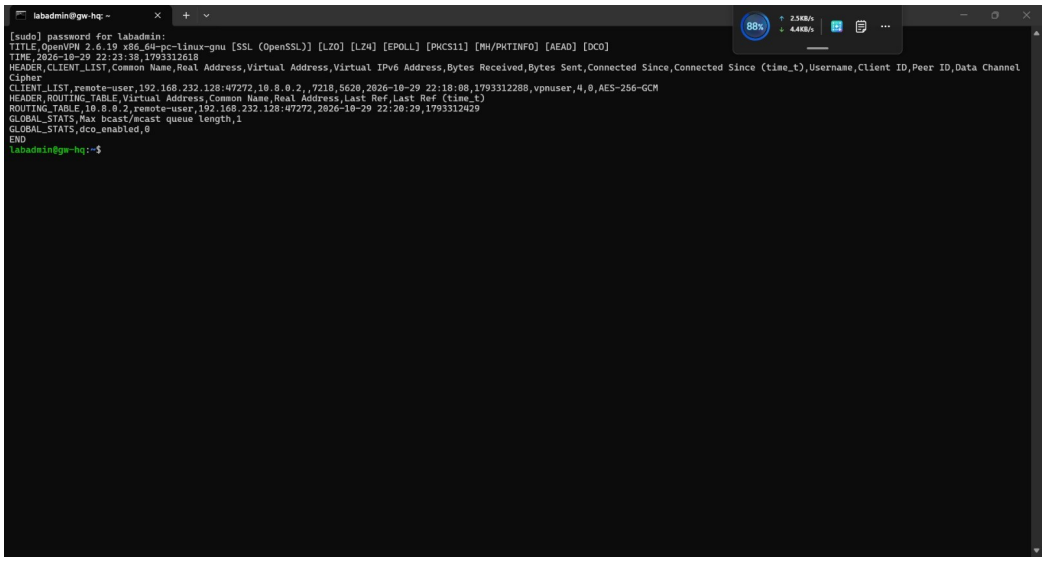


Figure 1. OpenVPN status file listing the connected client [E19]

3 Central logging design

gw-hq forwards openvpn*, sshd, sudo, charon*, vpn-test and kernel messages to gw-branch (192.168.232.132) over plain TCP port 514 with a disk-assisted queue [E27]. gw-branch stores them under /var/log/remote/gw-hq/ in separate files: charon-systemd, kernel, openvpn, sshd, sudo and vpn-test .log [E28]. A test message sent at 31 Oct 16:06:02 UTC arrived [E28].

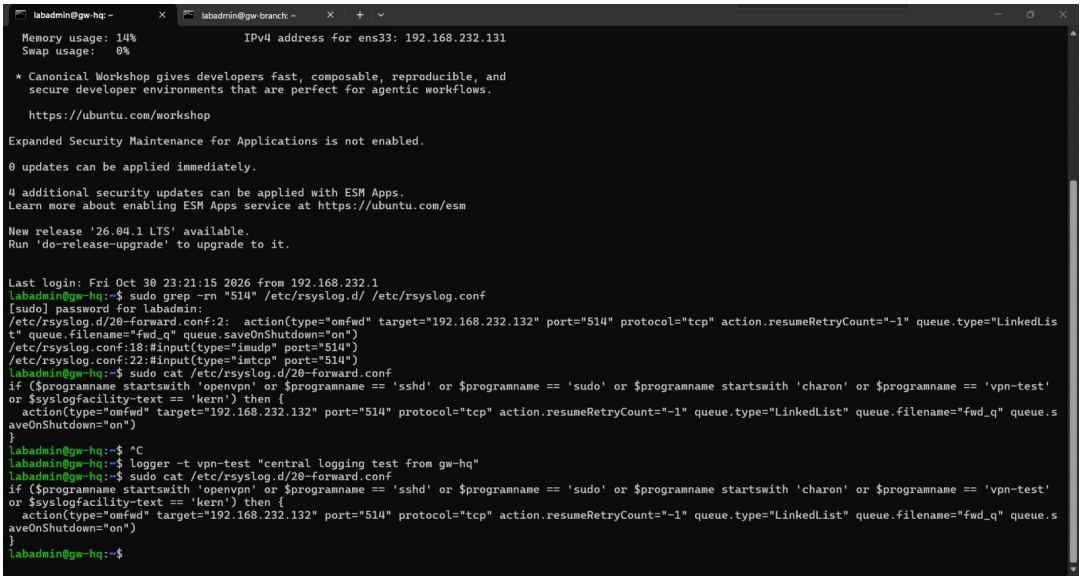


Figure 2. rsyslog forwarding rule on gw-hq [E27]

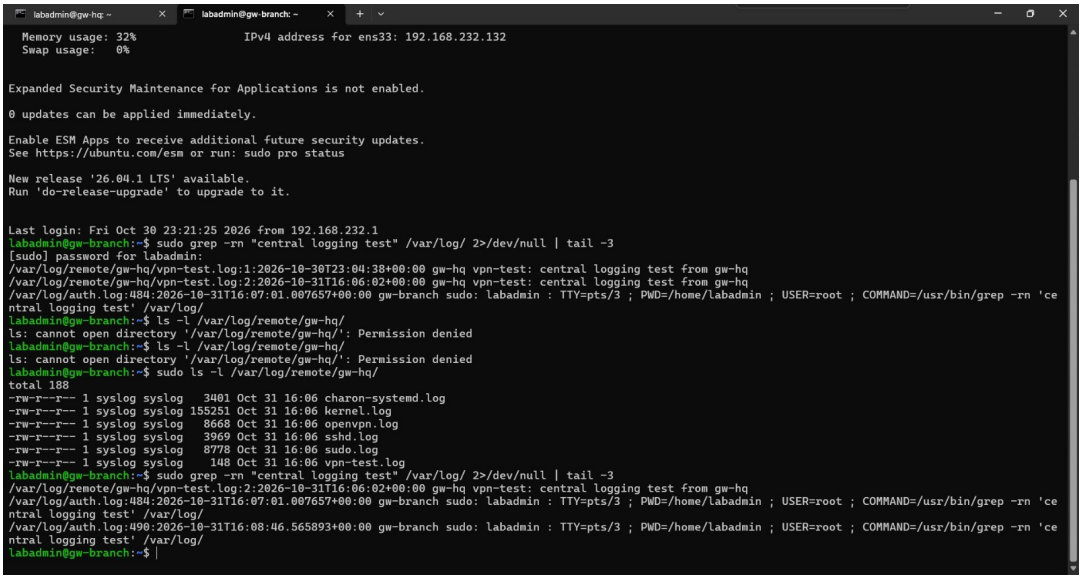


Figure 3. Test message received and per-service files on gw-branch [E28]

Weakness: TCP 514 has no TLS, so log content can be read or altered on the network between the two gateways.

4 rsyslog configuration

The sender rule on gw-hq is shown in [E27]. The receiver on gw-branch listens on TCP 514 and writes one file per service, as shown in [E28]. Full configuration files stay on the gateways.

5 Detection use cases

Use case	Indicator	Real example (UTC)	Evidence
Brute-force or wrong MFA code	"Invalid verification code"; AUTH_FAILED	23:55:36	[E08][E09]

Revoked certificate used	"certificate revoked"	From 23:40:48	[E24][E25]
Access to a forbidden network	Kernel log lines prefixed VPN-DENY	3 lines, 23:47:38-23:47:40	[E11]
Repeated SSH failures	fail2ban sshd jail: 3 failures from gw-branch, ban, then unban	See screenshot	[E17]
Who is connected	Client listed in status file	See screenshot	[E19]
Pipeline health	Test message from vpn-test arrives on gw-branch	31 Oct 16:06:02	[E28]

Exact fail2ban ban-time and retry values are not stated here; the ban and unban behaviour itself is shown in [E17].

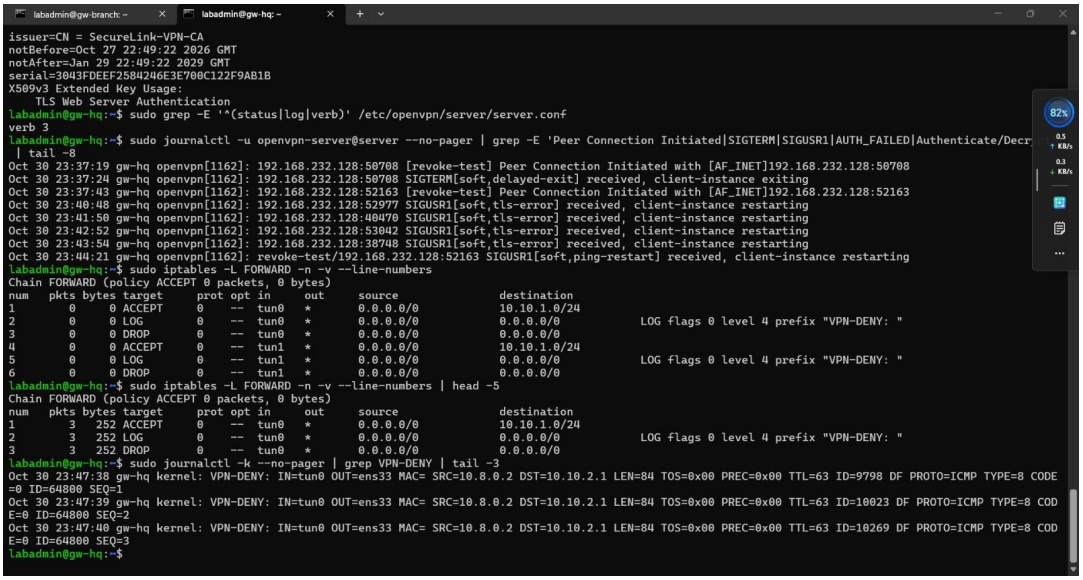


Figure 4. iptables rules, counters and VPN-DENY log lines [E11]

6 Limits and next steps

Logs can contain personal data such as user names and IP addresses (see D6). Time sync matters: server logs are UTC and the Kali client is UTC minus 4 hours.

Gap	Recommendation (not implemented)
No dashboards or alerting	Add alert rules for the use cases above; consider Zabbix or Nagios for uptime and ELK for search (assessed on paper only)
No retention settings	Define retention and rotation periods with the data protection officer
Plain TCP 514	Move to TLS-protected syslog
Single log server on the branch gateway	Use a separate, access-controlled log host

7 Implementation status

Category	Items
Implemented and verified	Local logs; VPN-DENY logging; central forwarding gw-hq to gw-branch; fail2ban sshd jail
Implemented but limited	Central logging without TLS; no dashboards, alerting or retention
Recommended, not implemented	TLS syslog; alerting; retention policy; monitoring platform